FROM DISTILLATION TO HARD NEGATIVE SAMPLING: REPRODUCTION & EXPERIMENTS ON SPLADE++

A Reproduction Study on Additive Techniques on SPLADE Models and an Experiment on Transfer-Learning

Based on the Study of Formal et al. (SIGIR '22)

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Information Retrieval 25/26
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PROJECT OBJECTIVES

Research Questions

- Can we validate the **additive improvements** of SPLADE++ training strategies (distillation, hard negative mining, pre-training) in a **zero-shot retrieval** context?
- How do different **training techniques** impact sparse retrieval **effectiveness**?
- How does the **zero-shot performance** of **sparse** retrievers compare to **dense** models on **unseen domains**?

Extension Experiment

- How does **fine-tuning** affect **zero-shot generalization**? Investigating domain adaptation in sparse retrieval models

EXPERIMENTAL SETUP OVERVIEW

BEIR Datasets (7 selected):

- SciFact - Scientific claim verification
- SCIDOCS - Scientific citation prediction
- FiQA - Financial QA
- NFCorpus - Medical information retrieval
- ArguAna - Argument retrieval
- TREC-COVID - COVID-19 research
- Webis-Touché-2020 - Argument search

Models Evaluated:

- BM25 (baseline)
- SPLADE v2 (original)
- SPLADE Distil (with knowledge distillation)
- SPLADE++ CoCondenser-SelfDistil
- SPLADE++ CoCondenser-EnsembleDistil
- Hybrid models (SPLADE++ + BM25)

Metrics used: NDCG@10, MRR@10, P@10, R@100, R@1K, Average FLOPS, Average Sparsity

THEORETICAL BACKGROUND: SPLADE V2

Sparse Expansion Mechanism

SPLADE projects input text into the **high-dimensional BERT WordPiece vocabulary** using the **Masked Language Modeling** (MLM) head . This performs implicit **expansion**, assigning weights to relevant terms even if they do not appear in the original text .

Key Architectural Improvement (v2)

While the original SPLADE used **Sum Pooling**, this version adopts **Max Pooling with log-saturation,** which extracts the strongest activation signal for each term across the sequence, reducing noise compared to summing all contributions.

Weighting Formulation

Token importance (*Wj*) is calculated via **log-saturation and max aggregation** over input tokens (*t*):

$$w_j = \max_{i \in t} \log(1 + \mathrm{ReLU} w_{ij}))$$

SPLADE V2 PERFORMANCE

Training Objective (Loss Function)

Training jointly optimizes ranking effectiveness and model efficiency by combining two components:

- **Ranking Loss** ($\mathcal{L}_{\text{InfoNCE}}$): Learns to distinguish relevant documents using **negative examples** mined from BM25 and the current batch.
- **Regularization** ($\mathcal{L}_{\text{FLOPS}}$): Penalizes dense representations to enforce the sparsity required for efficient indexing. The **lambda parameters** control the trade-off between effectiveness and sparsity.

$$\mathcal{L} = \mathcal{L}_{\text{InfoNCE, BM25}} + \lambda_q \mathcal{L}^q_{\text{FLOPS}} + \lambda_d \mathcal{L}^d_{\text{FLOPS}}$$

Dataset	BM25	SPLADE v2
ArguAna	0.4938	0.4206
FiQA	0.2259	**0.2932**
NFCorpus	0.313	**0.3259**
SCIDOCS	0.1534	0.1496
SciFact	0.6791	0.6405
TREC-COVID	0.6142	**0.646**
Touché-2020	0.3238	0.2026
Average	***0.4004***	***0.3826***

Performance comparison between **BM25** (baseline) and **SPLADE v2 checkpoint** on the BEIR datasets (NDCG@10). SPLADE outperforms BM25 in datasets requiring **semantic generalization** (e.g., TREC-COVID), but struggles significantly in **debate-oriented** domains like ArguAna.

KNOWLEDGE DISTILLATION FOR SPLADE

Knowledge Distillation (Theory)

- **Concept**: Training the sparse student model to mimic the scores of a powerful (but slow) **Cross-Encoder** teacher.
- **Loss Function**: Implementation of MarginMSE loss. The model minimizes the **Mean Squared Error** (MSE) between the teacher's margin (positive-negative gap) and the student's margin.
- **Impact**: Distillation is typically the **largest** single factor in performance improvement.

MarginMSE Loss:

$$\mathcal{L} = \mathcal{L}_{\text{MarginMSE, BM25}} + \lambda_q \mathcal{L}^q_{\text{FLOPS}} + \lambda_d \mathcal{L}^d_{\text{FLOPS}}$$

where $\mathcal{L}_{\text{Margin-MSE}} = (M_{\text{Teacher}} - M_{\text{Student}})^2$

SPLADE DISTIL PERFORMANCE

The data indicates that **SPLADE Distil checkpoint** is the **best-performing model** overall, achieving the highest average score of **0.4256 (NDCG@10)** and surpassing both the baseline and the standard **SPLADE v2**. Nevertheless, **BM25** remains competitive, outperforming both neural models by remaining more robust on **exact lexical matching** in argument retrieval and debate-oriented datasets, like ArguAna and Touché-2020.

Dataset	BM25	SPLADE v2	SPLADE Distil
ArguAna	0.4938	0.4206	0.4493
FiQA	0.2259	0.2932	**0.3413**
NFCorpus	0.313	0.3259	**0.3427**
SCIDOCS	0.1534	0.1496	**0.1551**
SciFact	0.6791	0.6405	**0.703**
TREC-COVID	0.6142	0.646	**0.7042**
Touché-2020	0.3238	0.2026	0.2833
Average	***0.4004***	***0.3826***	***0.4256***

HARD NEGATIVE MINING & PRE-TRAINING

Hard Negative Mining

Replaces standard BM25 negatives with **harder, high-scoring negatives** mined from stronger retrievers, to force the model to learn **finer-grained distinctions** between relevant and non-relevant documents.

- **Self-Mining**: Utilizes the model from the **previous training round** to retrieve **challenging negatives** for the current round, progressively exposing the model to harder examples as it improves.
- **Ensemble-Mining**: Aggregates negatives from **BM25 and a set of 12 various dense retrievers**, ensuring robustness against various retrieval failures.

Pre-training (CoCondenser)

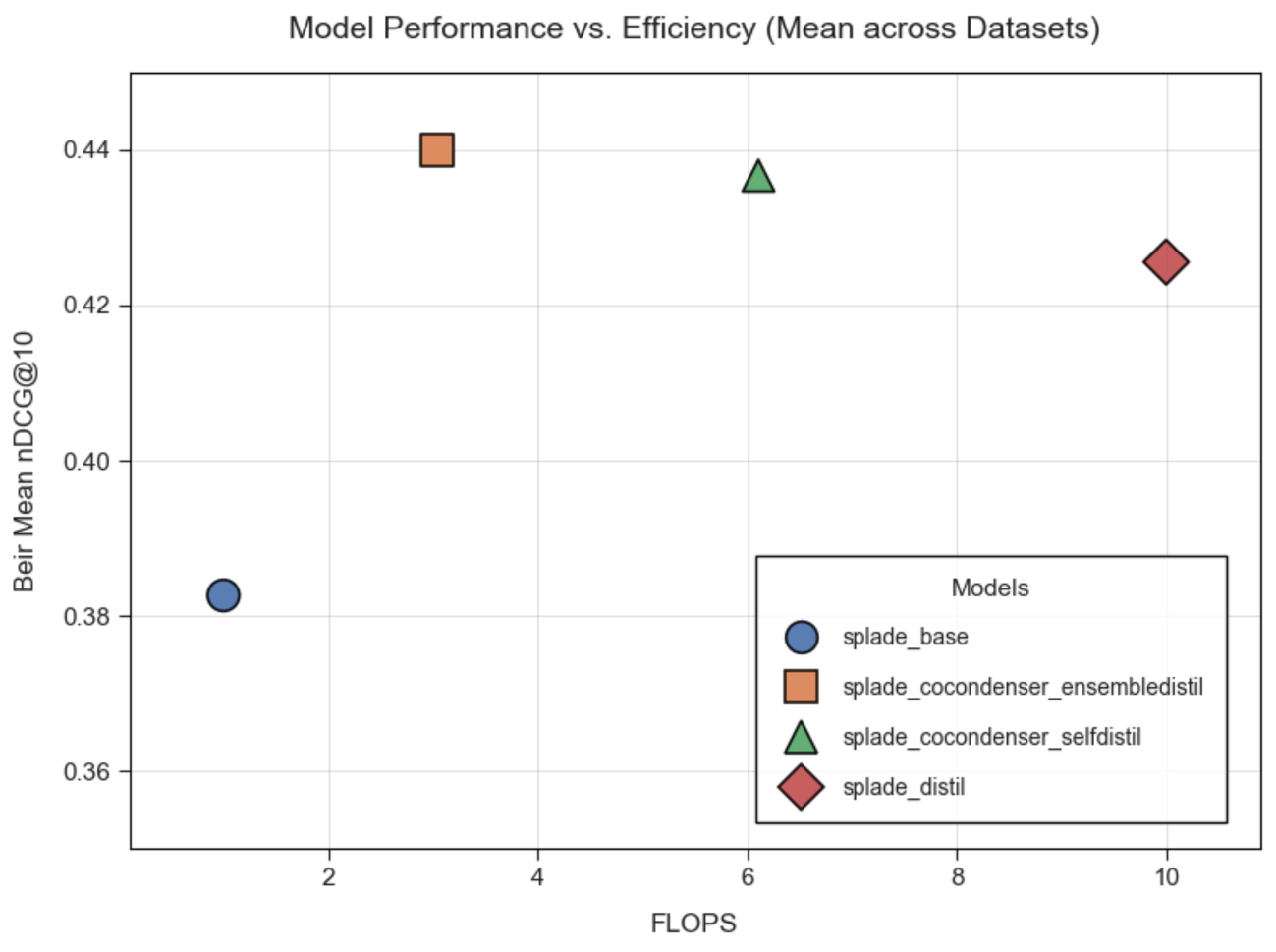
Initializing the model with **CoCondenser** instead of standard BERT. Unlike BERT, CoCondenser is a neural model pre-trained to capture **global text structure**, making it naturally better suited for ranking documents.

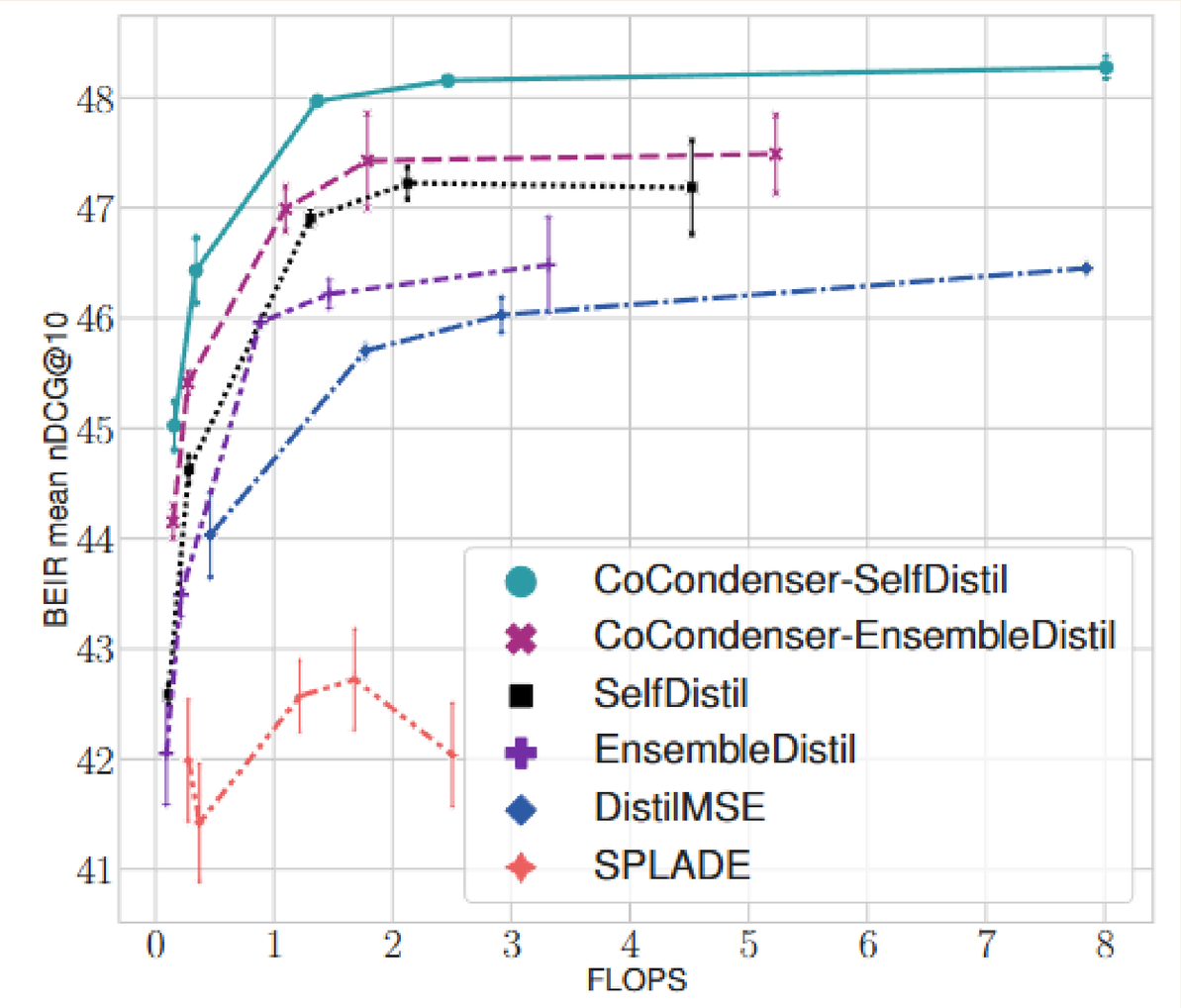
SPLADE++ MODELS PERFORMANCE

The results demonstrate a clear progression in performance with the addition of **advanced training strategies** (**SPLADE++ checkpoints**). While the base SPLADE model underperforms the BM25 baseline on average (0.3826 vs. 0.4004 NDCG@10), the introduction of **CoCondenser pre-training combined with Self and Ensemble Mining** yields the **highest overall effectiveness** (0.4367 and 0.44). Notably, the Ensemble configuration allows the neural model to finally **surpass** BM25 even on the challenging ArguAna dataset (0.5076 vs. 0.4938), overcoming a limitation observed in previous iterations.

Dataset	BM25	SPLADE v2	SPLADE Distil	SPLADE CoCondenser (Self-Distil)	SPLADE CoCondenser (Ensemble Distil)
ArguAna	0.4938	0.4206	0.4493	0.4926	**0.5076**
FiQA	0.2259	0.2932	0.3413	**0.3617**	**0.3559**
NFCorpus	0.313	0.3259	0.3427	**0.3545**	**0.3525**
SCIDOCS	0.1534	0.1496	0.1551	**0.1604**	**0.1581**
SciFact	0.6791	0.6405	0.703	**0.7152**	**0.7079**
TREC-COVID	0.6142	0.646	0.7042	**0.7246**	**0.7254**
Touché-2020	0.3238	0.2026	0.2833	0.2479	0.2726
Average	***0.4004***	***0.3826***	***0.4256***	***0.4367***	***0.44***

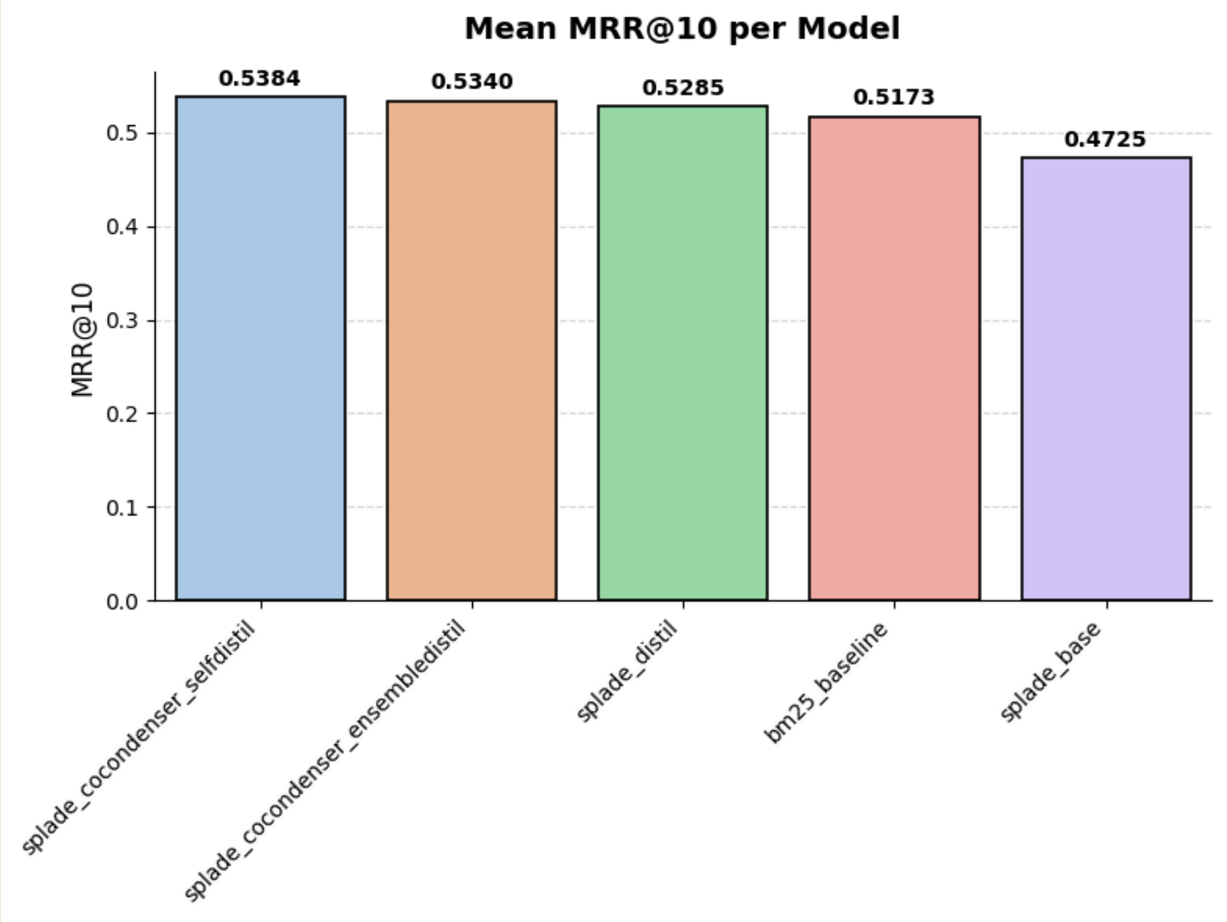
COMPARISON WITH OFFICIAL BASELINES

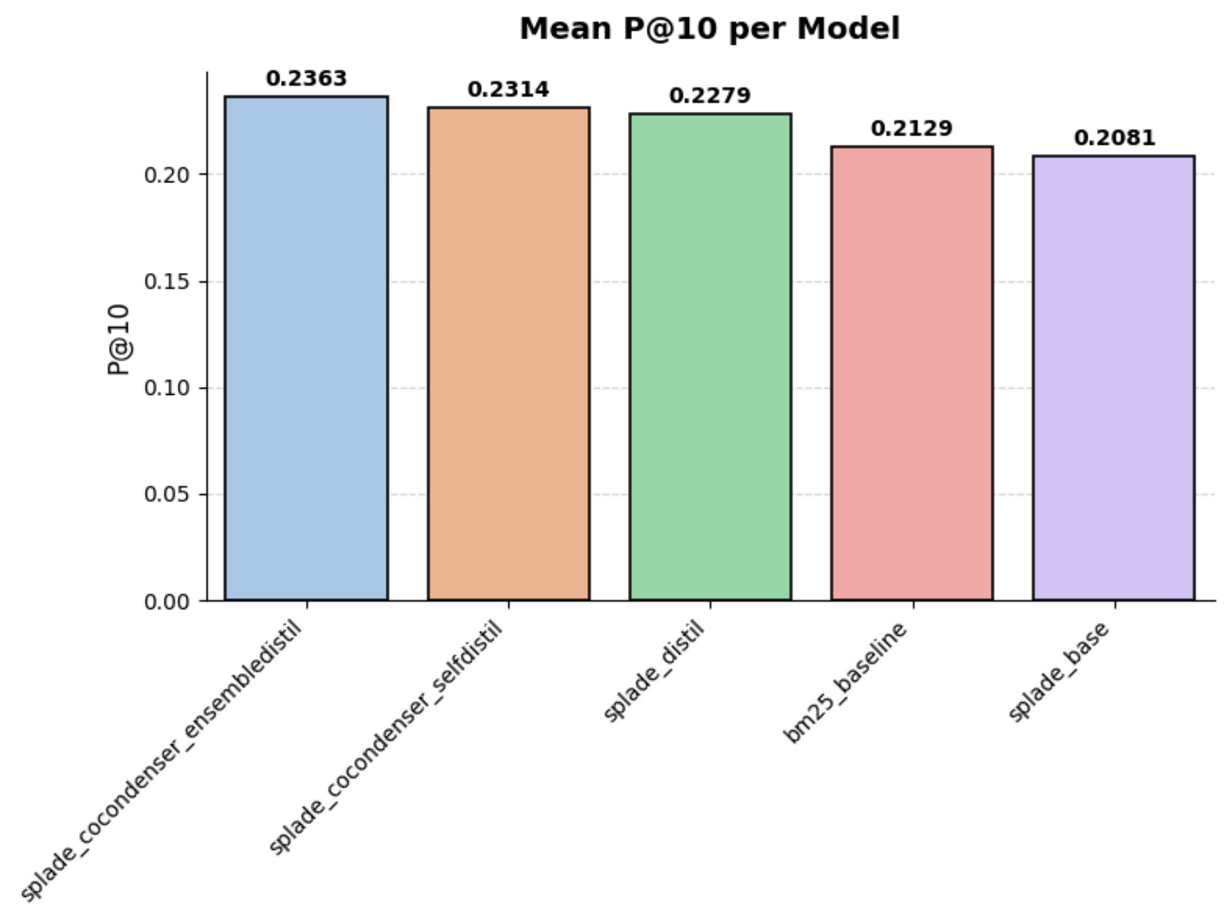




The inference results exhibit high fidelity to the official figures reported in the **original paper (Formal et al.).** The observed performance trends are consistent, confirming that SPLADE++ models consistently outperforms previous iterations, with score discrepancies remaining within negligible margins expected from **hardware/software variations**.

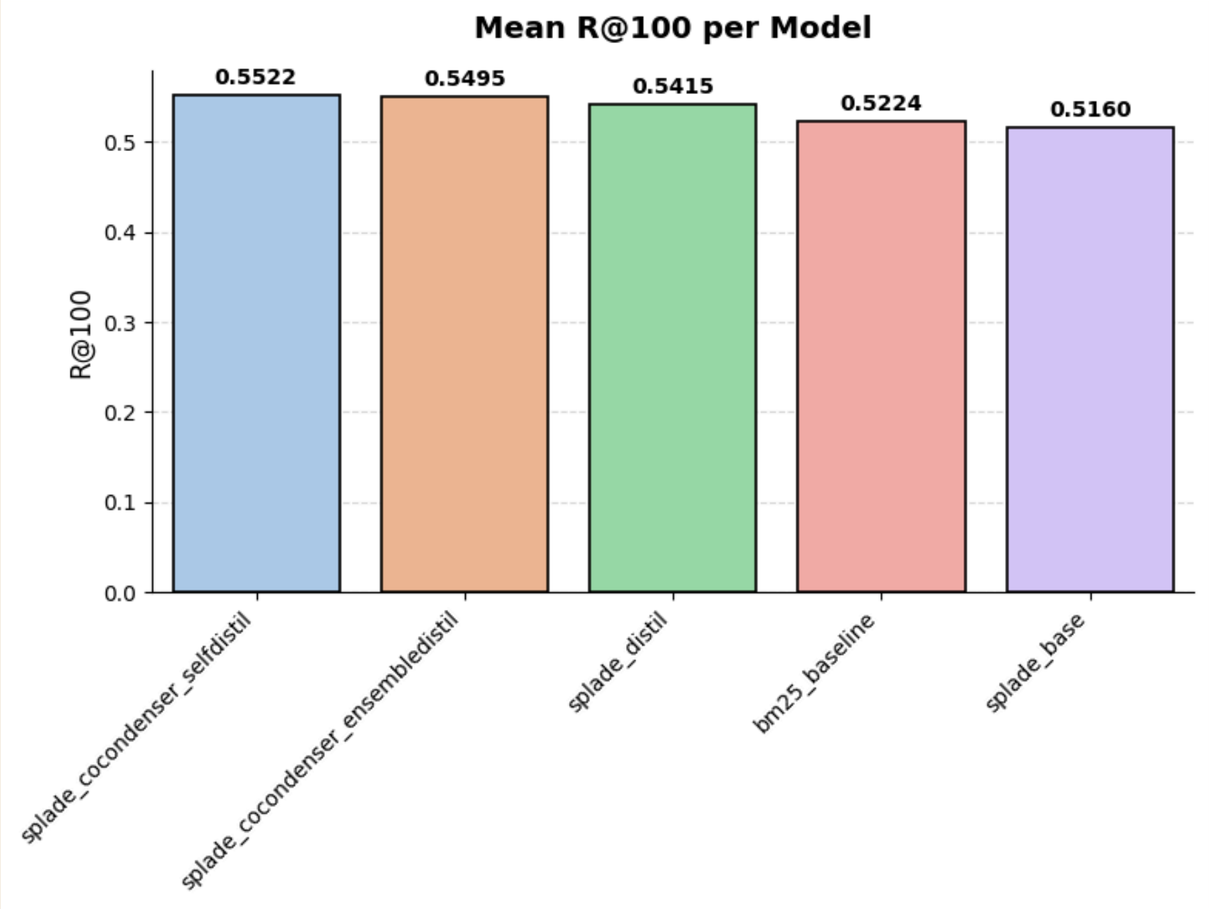
MODEL PERFORMANCE COMPARISON (1)

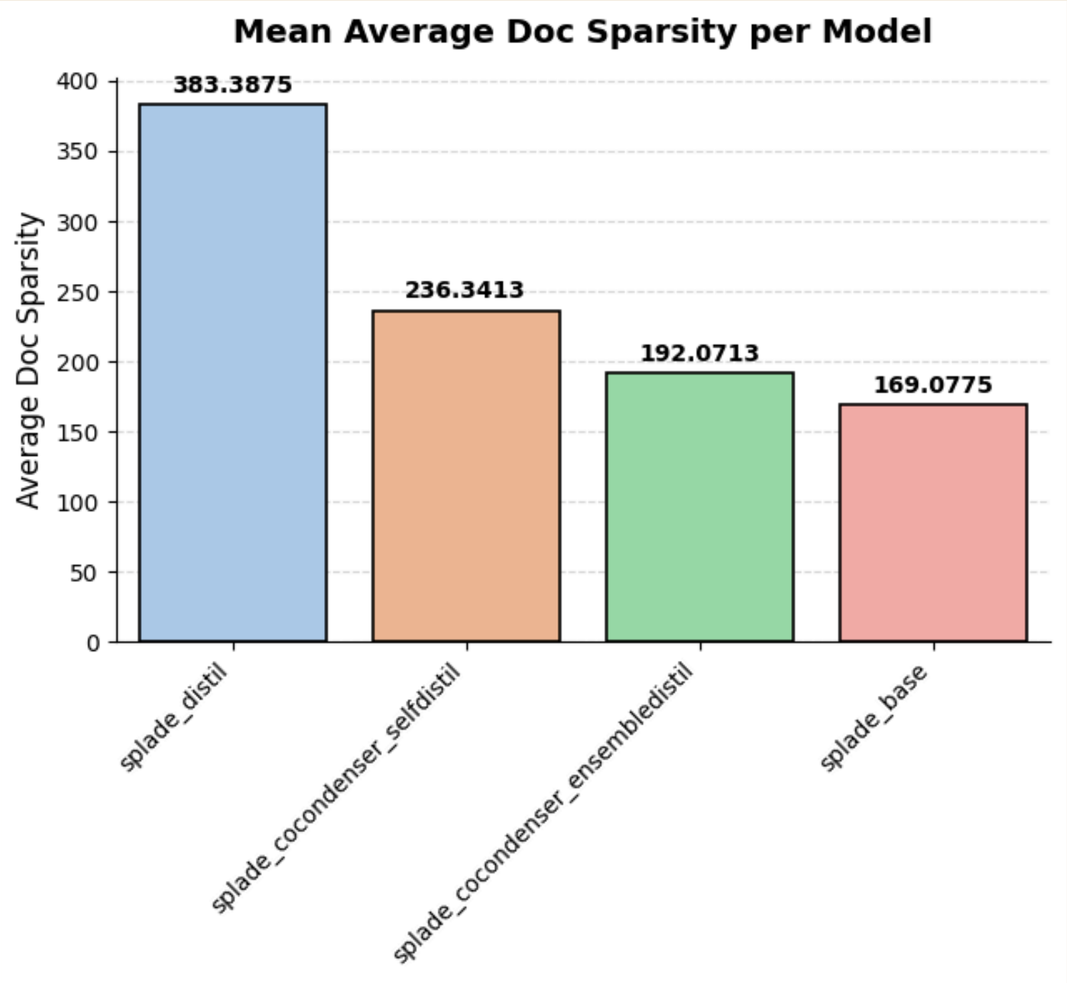




The comparison reveals metric-dependent performance patterns: **CoCondenser-SelfDistil** leads in **MRR@10 (0.5384)**, while **CoCondenser-EnsembleDistil** excels in **P@10 (0.2363)**. Simpler SPLADE variants show lower scores, with overall variance remaining narrow (±12-13%), indicating robust performance across all SPLADE configurations.

MODEL PERFORMANCE COMPARISON (2)





SPLADE models consistently **outperform** BM25 in **Recall@100**, with **CoCondenser** variants achieving the **best results** thanks to advanced training strategies. **Distillation** increases document density, resulting in **higher sparsity values,** while **CoCondenser-based models** produce **more compact** sparse representations. This highlights a clear effectiveness–efficiency trade-off, where stronger recall usually requires **denser** document representations.

HYBRID MODELS

Why go Hybrid?

Neural models excel at **semantics**, while BM25 is unbeatable at **exact lexical matching**. **Combining** them covers "blind spots" on both sides.

- Goal: Enhance **robustness**, particularly in **Zero-Shot scenarios** where the neural model might drift due to domain shift.

Method: Score Fusion (Linear Combination)

Formula:

$$S_{\text{Hybrid}} = (1 - \alpha) \cdot S_{\text{SPLADE}} + \alpha \cdot S_{\text{BM25}}$$

Implementation Code:

1. **Min-Max Normalization**: Both SPLADE and BM25 scores are normalized to the [0, 1] range to make them comparable;
2. **Weighted Sum**: Scores are fused using a scalar α (alpha).

The efficiency is **post-hoc fusion.** No re-training or heavy computation required.

HYBRID MODELS PERFORMANCE

The data confirms that the **Hybrid Strategy (SPLADE + BM25)** consistently enhances overall performance, raising the average nDCG@10 for both models (e.g., Ensemble improves from 0.4400 to 0.4532). The benefit is most dramatic in **debate-oriented tasks like Touché-2020**, where the hybrid approach **boosts score in Self-Distil** by over 41% (from ~0.24 to ~0.34), effectively correcting the neural model's tendency to drift in zero-shot argument retrieval scenarios. However, in semantic-heavy tasks like FiQA, the **pure neural model remains superior,** suggesting that lexical interpolation should be tuned per domain.

Dataset	Self-Distil	Self-Distil + BM25	Ensemble Distil	Ensemble Distil + BM25
ArguAna	0.4926	**0.5179**	0.5076	**0.5329**
FiQA	0.3617	0.3164	0.3559	0.3154
NFCorpus	0.3545	0.3493	0.3525	0.3479
SCIDOCS	0.1604	**0.168**	0.1581	**0.1662**
SciFact	0.7152	**0.7217**	0.7079	**0.7202**
TREC-COVID	0.7246	**0.7437**	0.7254	**0.7441**
Touché-2020	0.2479	**0.3455**	0.2726	**0.3454**
Average	***0.4367***	***0.4518***	***0.44***	***0.4532***

	SPLADE++‡ + BM25	SPLADE++† + BM25
nDCG@10	52.1	52.1

Reference (Formal et al.), Mean NDCG@10.

SOTA COMPARISON

Dataset	BM25	TAS-B	Contriever	ColBERTv2	SPLADE++ Self (Formal et al.)	SPLADE++ Ens (Formal et al.)	SPLADE++ Self (Mine)	SPLADE++ Ens (Mine)
ArguAna	31.5	42.9	44.6	46.3	51.8	**52.5**	49.3	50.8
FiQA-2018	23.6	30	32.9	35.6	34.9	34.8	**36.2**	35.6
NFCorpus	32.5	31.9	32.8	33.8	34.5	34.8	**35.5**	35.2
SCIDOCS	15.8	14.9	**16.5**	15.4	16.1	15.9	16	15.8
SciFact	66.5	64.3	67.7	69.3	71	70.2	**71.5**	70.8
TREC-COVID	65.6	48.1	59.6	**73.8**	72.5	72.7	72.5	72.5
Touché-2020	**36.7**	16.2	23	26.3	24.2	24.5	24.8	27.3

The table validates the accuracy of my **inference and evaluation pipeline.** By benchmarking the **official pre-trained checkpoints ('Mine')** against the figures **reported by Formal et al. (2022)**, we observe a **high degree of fidelity.** This confirms that my setup effectively **replicates** state-of-the-art performance, matching the original paper's scores on the datasets while consistently **outperforming dense baselines** such as TAS-B and Contriever.

FINE-TUNING EXPERIMENT

Motivation: Domain Adaptation & Generalization

SCIDOCS (Citation Recommendation) performance is **suboptimal (~0.16 nDCG@10)**, significantly lagging behind other benchmarks. The **proposed solution** is a fine-tuning on **SciFact** (Scientific Fact Checking), a related scientific evidence retrieval task, can induce **positive transfer learning**, boosting SCIDOCS performance through **shared domain** semantics.

Research Question

Can we enhance domain-specific effectiveness without catastrophic forgetting?
Control Metric: We monitor the dataset **FiQA** (Financial QA) to ensure the model retains its ability to **generalize across unrelated domains** (e.g., finance) while **specializing in science.**

FINE-TUNING EXPERIMENT SETUP

Core Configuration

- **Base Model:** SPLADE++ (CoCondenser Self-Distil), which performed **the best** on the BEIR datasets.
- **Objective:** Assess transfer learning from **SciFact** (Source) to **SCIDOCS** (Target) while monitoring **FiQA** for potential overfitting.

Regularization Strategy

Two distinct configurations were evaluated to analyze the trade-off between **semantic expressiveness and retrieval efficiency (λ_{FLOPS}):**

Feature	Configuration A (Dense)	Configuration B (Sparse)
Regularization	Lambda = 0.0008	Lambda = 0.008
Focus	**High Expressiveness**	**High Efficiency**
Sparsity	Low (More active terms)	High (Stricter constraints)
Trade-off	Higher index size & latency	Faster retrieval & lower storage
Risk	Computational overhead	Underfitting complex semantics

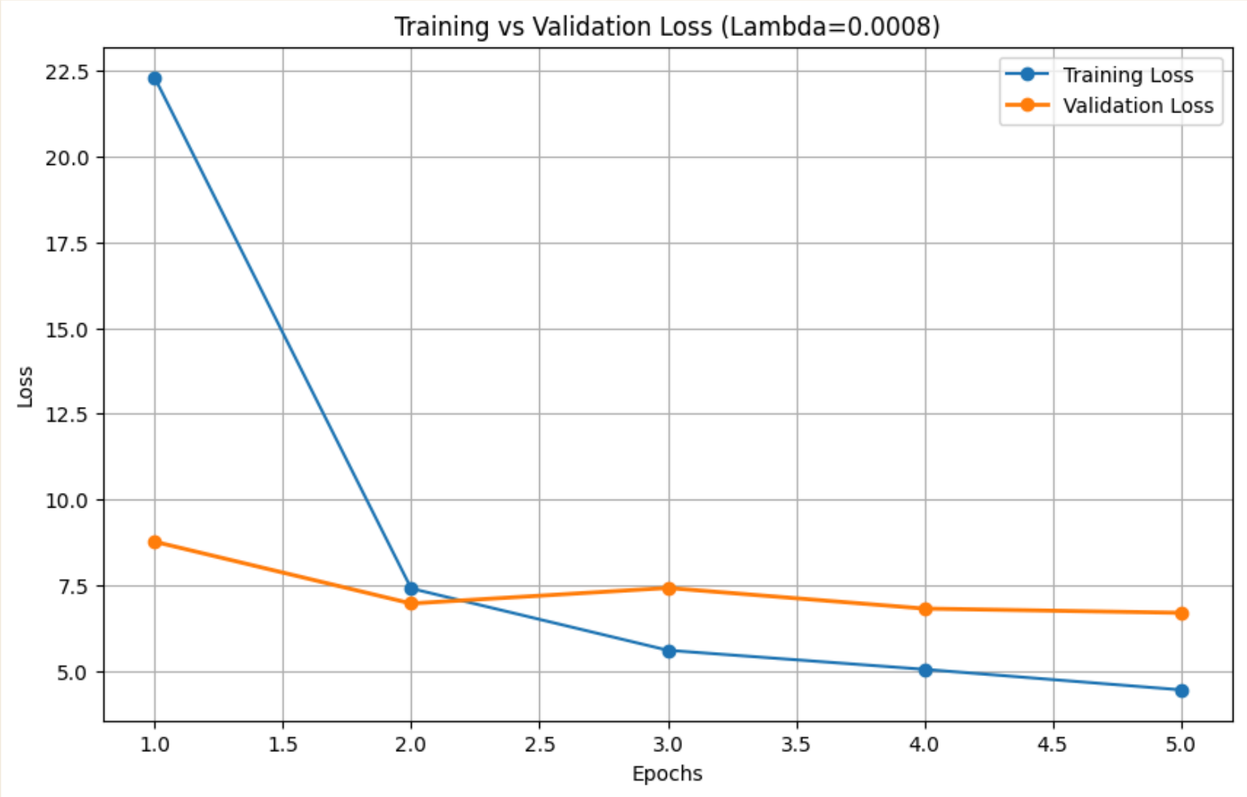
TRAINING DYNAMICS (1)

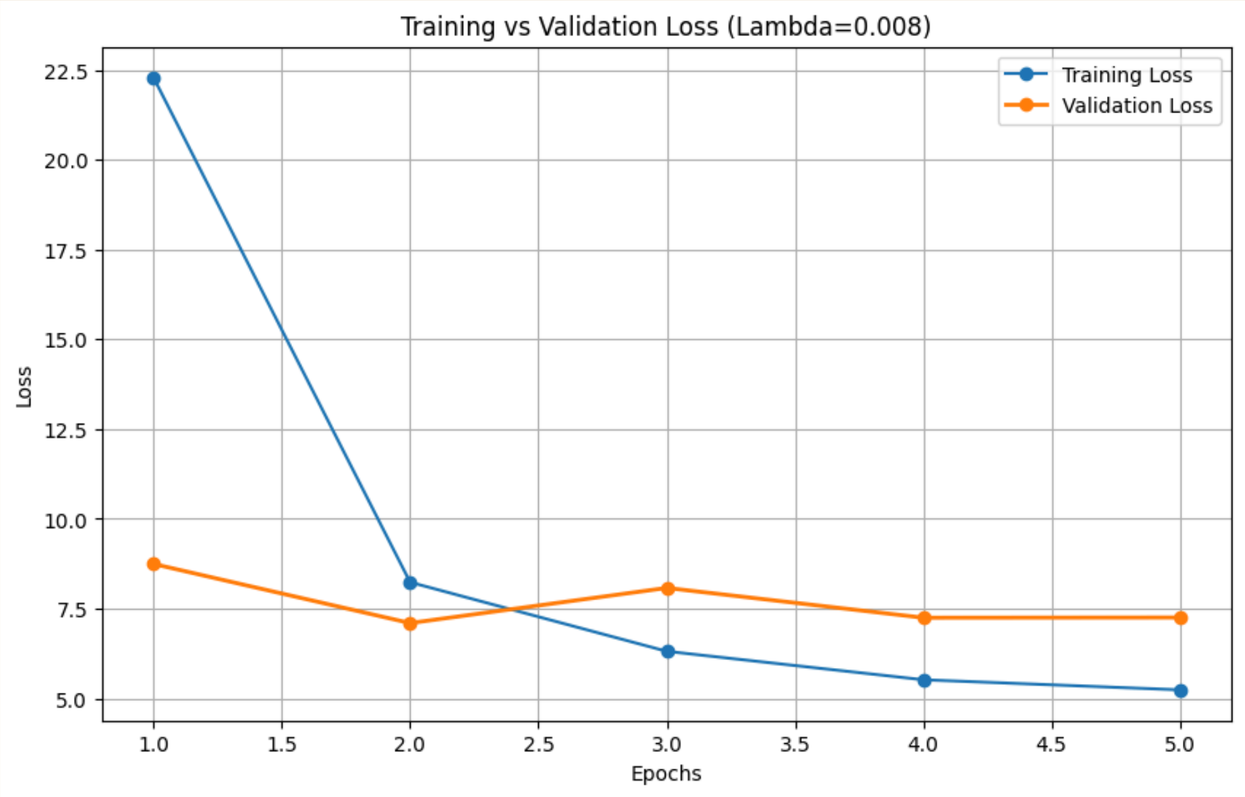
Training Protocol (Methodology)

Both models were fine-tuned using a consistent hyperparameter regime to ensure comparable results:

- **Duration:** Maximum of **5 Epochs** with **Early Stopping (Patience = 3)** to prevent **overfitting**.
- **Optimization: AdamW optimizer** with **Gradient Accumulation** (Batch Size: 2 x 16 steps) to simulate larger batches on limited hardware.
- **Learning Rate Strategy:** An **very low** Learning Rate (lr = 5e-6) was applied to mitigate catastrophic forgetting. This specific configuration proved to be the **most effective** for maximizing model performance.
- **Loss Function:** MarginMSELoss combined with FLOPS **Regularization** (λ).

TRAINING DYNAMICS (2)





Both regularization settings show **similar training dynamics**: **training loss drops rapidly** from 22.5 to ~4.5 over 5 epochs, while **validation loss remains stable** at 8.5-6.5. The consistent train-validation gap in both λ=0.0008 (dense) and λ=0.008 (sparse) configurations indicates **comparable convergence** behavior with **no signs of overfitting.**

FINE-TUNING RESULTS (1)

Key Observations

- **Stability vs. Gain:** The **Sparse model (Lambda = 0.008**) proved to be robust, as the most effective configuration. It successfully balanced the **trade-off between efficiency and effectiveness**, achieving a marginal performance gain **(+0.1 NDCG@10)** on the target domain (**SCIDOCS**) while effectively **reducing** the index size.
- **Negative Transfer:** A slight **regression** was observed on the **source dataset (SciFact, -0.8 to -1.5 points)** and the **out-of-domain benchmark (FiQA, -1.1/-1.2 points)**, suggesting that the fine-tuning regime might have disrupted the pre-trained weights without sufficiently capturing the new domain semantics.

Dataset Context	Baseline (No FT)	FT Dense (λ=0.0008)	FT Sparse (λ=0.008)
SciFact (Train)	71.5	**70.8**	**70**
SCIDOCS (Target)	16	16	**16.1**
FiQA (General)	36.2	**35.1**	**35**

The table summarizes the experimental results of fine-tuning SPLADE++ on the SciFact dataset. Contrary to the initial hypothesis, the domain adaptation process yielded neutral results on the target task (SCIDOCS), where performance remained stagnant (~16.0) across all configurations and worsened on other datasets.

FINE-TUNING RESULTS (2)

Fragility of Sparse Expansions

- The minimal gains on the target domain suggest that the base **SPLADE++** model (pre-trained on MS MARCO) had already reached a **performance saturation** point for this task. SPLADE is designed as a **robust zero-shot retriever**. Its learned term expansion logic is **highly generalized**; fine-tuning on a small dataset (SciFact, ~5k documents) likely offers **diminishing returns** compared to the massive pre-training, acting more as **noise** than signal.

- The fine-tuning process likely **biased the model towards dataset-specific artifacts** (SciFact-specific terms) rather than **learning transferable scientific concepts**, causing the model to lose the "**universal**" matching capability it acquired during pre-training. **Sparse neural retrievers** like SPLADE may require **significantly larger corpora** to benefit from transfer learning. For low-resource scenarios, the zero-shot configuration proves to be **superior and more reliable** than few-shot fine-tuning strategies.

CONCLUSIONS

Summary of Findings

- **Validation Success:** Validated the SPLADE++ training pipeline. My results match the **official SOTA baselines** (e.g., TREC-COVID: 72.5), confirming the **robustness** of the Distillation, Hard Negative Mining and CoCondenser initialization **techniques**.
- **Zero-Shot Superiority:** Sparse neural retrievers proved **more effective** than dense baselines (TAS-B, Contriever) in **zero-shot settings**, offering a better **balance of efficiency and lexical interpretability.**
- **Limits of Fine-Tuning:** The domain adaptation experiment on SciFact revealed the **"Stability-Plasticity" dilemma**. The strong pre-training resisted adaptation. The model **overfitted** to the source data's quirks rather than learning **transferable semantic features**.
- **Future Work:** Investigate transfer learning using **larger, intermediate scientific corpora** to prevent overfitting.

References

Formal et al. (2022): From Distillation to Hard Negative Sampling: Making Sparse Neural IR Models More Effective. (SIGIR '22).
Thakur et al. (2021): BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. (NeurIPS '21).

THANK YOU